Inventory Management System - Lab 8 UML Class Diagram

PhysicalProduct
- __weight : float
+ BASE_SHIPPING : float = 5.00
+ RATE_PER_KG : float = 2.50
+ get_weight() : float
+ display_information() : void
+ to_dict() : dict
+ get_fulfilment_info() : str
+ calculate_shipping_cost() : float
+ from_dict(data : dict, supplier : Supplier) : PhysicalProduct

«interface»
CanBeShipped
+ *calculate_shipping_cost() : float*

DigitalProduct
- __download_size : float
+ get_download_size() : float
+ display_information() : void
+ to_dict() : dict
+ get_fulfilment_info() : str
+ from_dict(data : dict, supplier : Supplier) : DigitalProduct

SubscriptionProduct
- __billing_months : int
+ get_billing_months() : int
+ display_information() : void
+ to_dict() : dict
+ get_fulfilment_info() : str
+ from_dict(data : dict, supplier : Supplier) : SubscriptionProduct

«abstract»
Product
- __product_id : str
- __name : str
- __price : float
- __stock : int
- __supplier : Supplier
+ Product(product_id : str, name : str, price : float, stock : int, supplier : Supplier)
+ get_product_id() : str
+ get_name() : str
+ get_price() : float
+ get_stock() : int
+ get_supplier() : Supplier
+ update_stock(quantity : int) : bool
+ display_information() : void
+ to_dict() : dict
+ *get_fulfilment_info() : str*

Supplier
- __supplier_id : str
- __name : str
- __contact_email : str
+ get_supplier_id() : str
+ get_name() : str
+ get_contact_email() : str
+ display_information() : void
+ to_dict() : dict
+ from_dict(data : dict) : Supplier

1 — supplied by — 0..*

Inventory
- __name : str
- __products : Product [0..*]
+ add_product(product : Product) : bool
+ remove_product(product_id : str) : Product [0..1]
+ transfer_product(product_id : str, target : Inventory) : bool
+ find_product(product_id : str) : Product [0..1]
+ search_product(keyword : str) : Product [0..*]
+ clear() : void
+ display_all_products() : void
+ get_products() : Product [0..*]

0..* — contains — 0..1

1 — refers to — 0..*

Order
- __order_id : str
- __order_lines : OrderLine [0..*]
+ get_order_id() : str
+ add_order_line(product : Product, quantity : int) : void
+ remove_order_line(product_id : str) : bool
+ get_order_lines() : OrderLine [0..*]
+ calculate_total() : float
+ display_information() : void

1 — owns — 0..*

OrderLine
- __product : Product
- __quantity : int
+ get_product() : Product
+ get_quantity() : int
+ calculate_subtotal() : float
+ display_information() : void

Legend
- private + public *italic* = abstract
solid line + hollow triangle = inheritance (is-a)
dashed line + hollow triangle = interface realisation
plain solid line = association (no ownership)
hollow diamond = aggregation (whole end; parts independent)
filled diamond = composition (whole end; owns part lifecycle)
Type [0..*] = collection, Type [0..1] = may be absent (None)